



THE UNIVERSITY OF
MELBOURNE

ACCT90002

Financial Statement

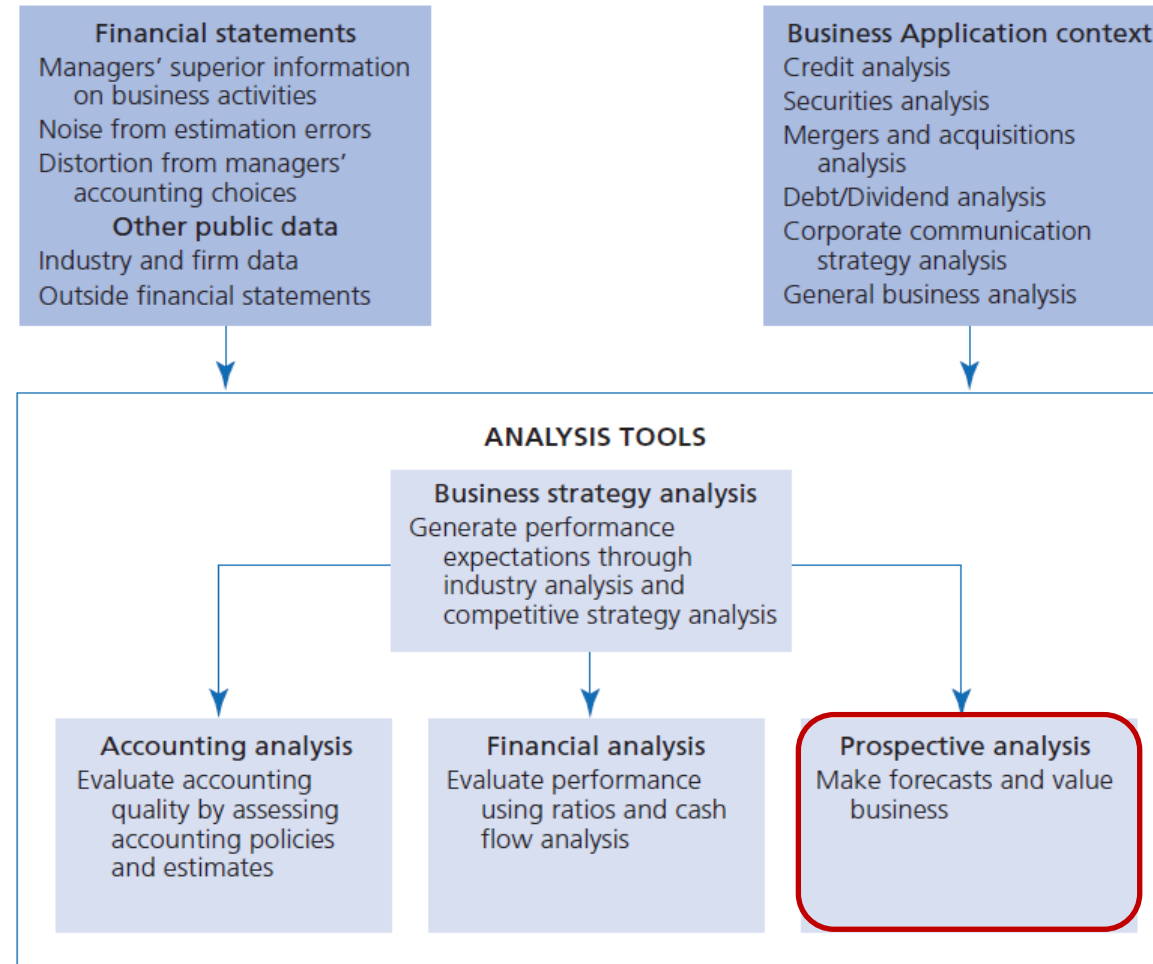
Analysis

Week 9: Valuation (1)

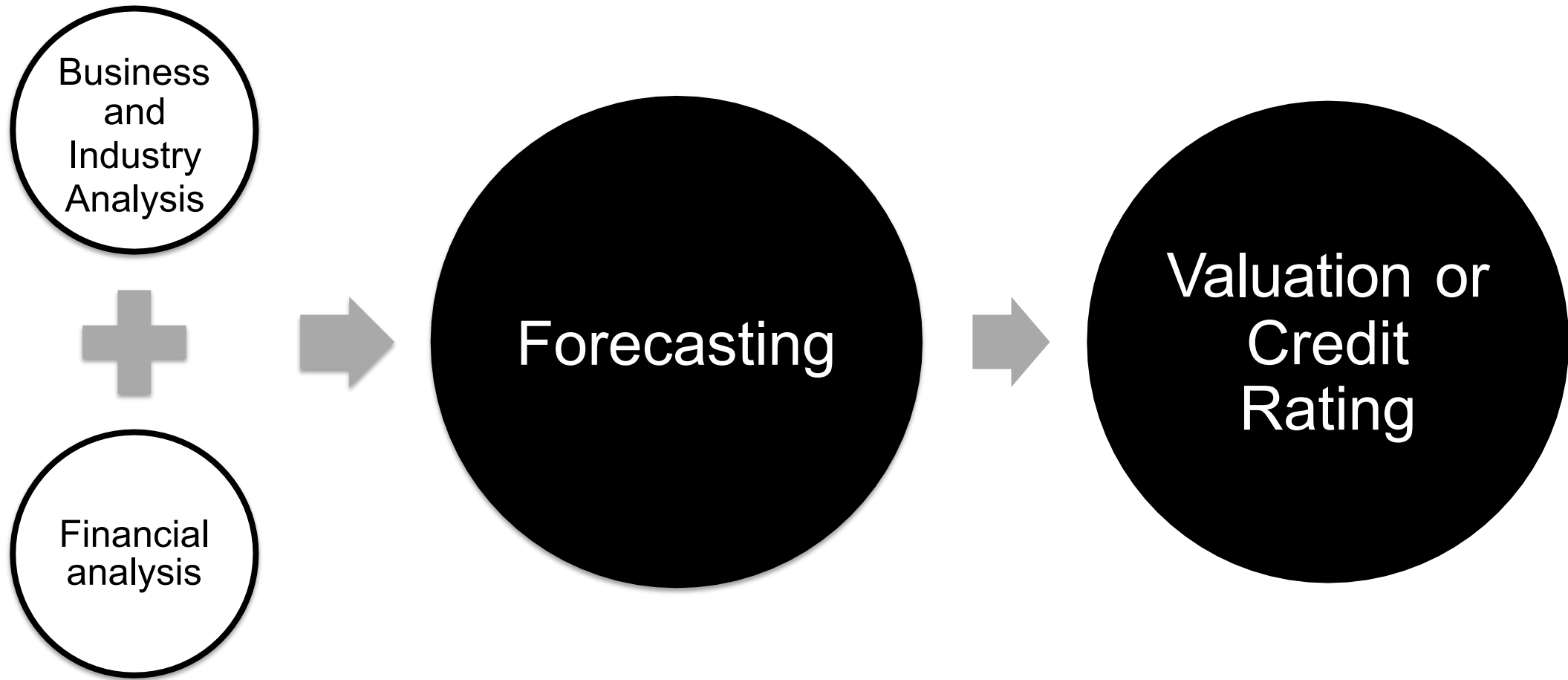


Framework - Financial Statement Analysis

FIGURE 1.3 Analysis using financial statements



Role of forecasting in user value chain



Fundamental valuation

- **Fundamental analysis** to estimate the value of a company
 - Used by equity investors, lenders, analysts, etc.
- This **valuation approach** uses basic **accounting measures** or **fundamentals**

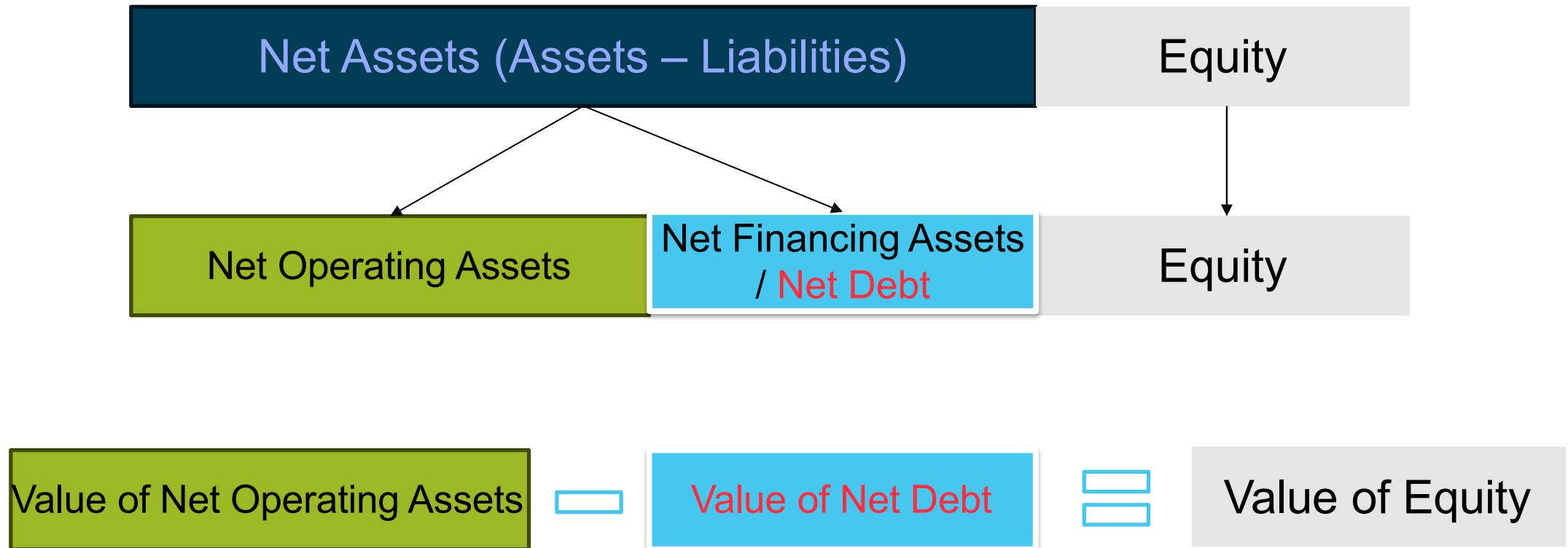


Equity valuation

- Our focus will be on the valuation of equities.
- It is important to conceptualise the company when we decide on an approach to valuation.
- Two potential ways to analyze a company:
 - Value the Equity
 - Value the [Equity] in the form of **Value the Operating Assets less Value of the Debt**

Equity valuation

- Remember from last week:





Equity valuation

- **Enterprise value =** Value of Net Operating Assets
 - Reflects the value of the operating components of the business
 - Should be computed using “fair” or market values
 - Often uses book (accounting) value of net debt when no ready access to market or fair values
- **Market capitalisation/Equity value =** Value of Equity
 - Total value of all the shares outstanding

Directly valuing Equity

Cash Flows / Earnings

- Related to equity holdings

eg. Free Cash Flow to
Equity, Dividends, Net
Profit

Discount Rate

- (Expected) returns to
shareholders

Value of Net Operating Assets



Value of Net Debt



Value of Equity



Valuing NOA and Debt

Cash Flows / Earnings

- Related to the NOA
- eg. Free Cash Flow ex
Interest, NOPAT

Discount Rate

- (Expected) returns to all funders of the NOA
- What will this include?

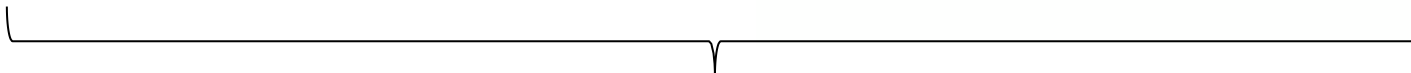
Value of Net Operating Assets



Value of Net Debt



Value of Equity



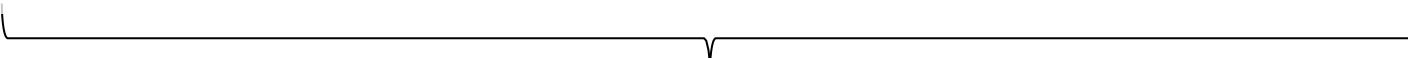
Valuing NOA and Debt

Cash Flows / Earnings

- Related to net debt
eg. Interest, coupons

Discount Rate

- (Expected) returns to all funders of the Debt
- Cost of debt (market determination of specific interest rates)

$$\text{Value of Net Operating Assets} - \text{Value of Net Debt} = \text{Value of Equity}$$


What are these fundamentals we use





Valuation models

The value of a share is the present value of future dividends.

Three methods are derived from this rule:

1. Discounted dividends
2. Discounted cash flows
3. Abnormal earnings

Each of these methods focuses the analyst's attention on different underlying variables and requires a different level of structure.

Intrinsic valuation



Valuation models—cont'd

No one method is superior to the others.

Using the same assumptions about firm fundamentals should yield the same value estimates from any of the methods used.

Methods can differ in the following aspects:

- Focus – earnings or cash flow
- Amount of analysis or structure required
- Terminal value implications

Common fundamental valuation models

1. Discounted dividends:

- Value of equity = present value (PV) of forecasted **future dividends**

2. Discounted cash flow:

- Value of equity = PV of multiple year forecasts of **free cash flows to equity claim holders**

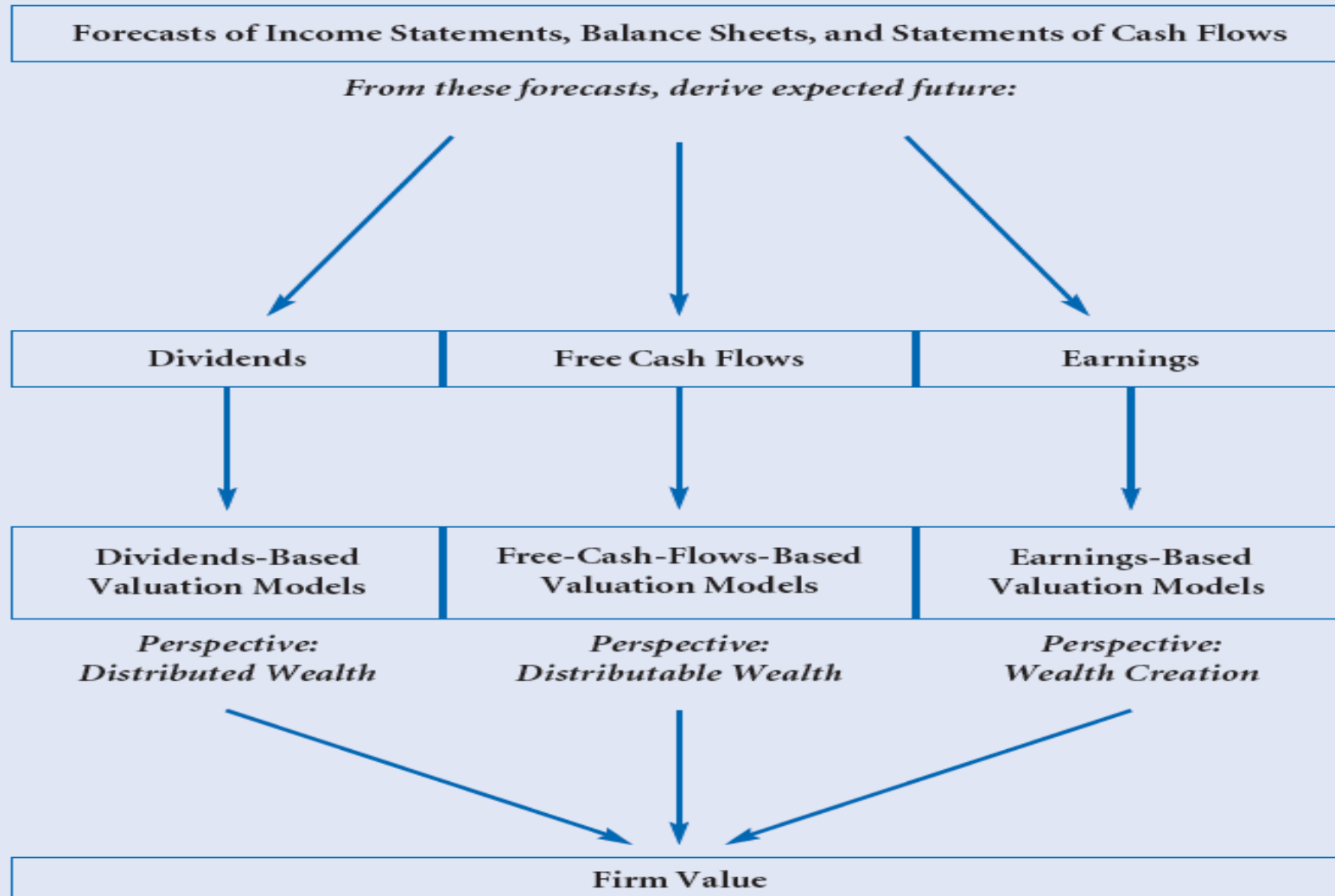
3. Discounted abnormal earnings (residual income):

- Value of equity = sum of **equity book value** and discounted forecasts of **abnormal earnings**

In principle – all three formulae give the same firm value **Focus on long term**

Approaches to firm valuation

Conceptual Illustration of Equivalent Approaches to Valuation Using Dividends, Cash Flows, and Earnings



Basic steps of fundamental valuation

Valuation involves three basic steps:

1. Forecasting **future values** of some **financial attributes** (i.e., **fundamentals**):
 - distributable or free cash flows
 - accounting earnings
 - balance sheet book values
2. Determining the **risk** or **uncertainty** *associated* with the attribute's forecasted future value [i.e., ***discount rate***]
3. Determining the **discounted/present value** of the expected future values of the value-relevant attribute
 - the ***discount rate*** reflects the **risk or uncertainty** inherent in the value attribute

Discounted Dividend Model (DDM)

Equity Value = PV of future dividends (including any liquidating dividend)

$$Equity_Value = \frac{DIV_1}{(1+r_e)} + \frac{DIV_2}{(1+r_e)^2} + \frac{DIV_3}{(1+r_e)^3} + \dots$$

Assuming **constant** dividend growth rate (g_d) *indefinitely*:

$$Equity_Value = \frac{DIV_1}{(r_e - g_d)}$$

Cost of equity capital



Determining Cost of Capital

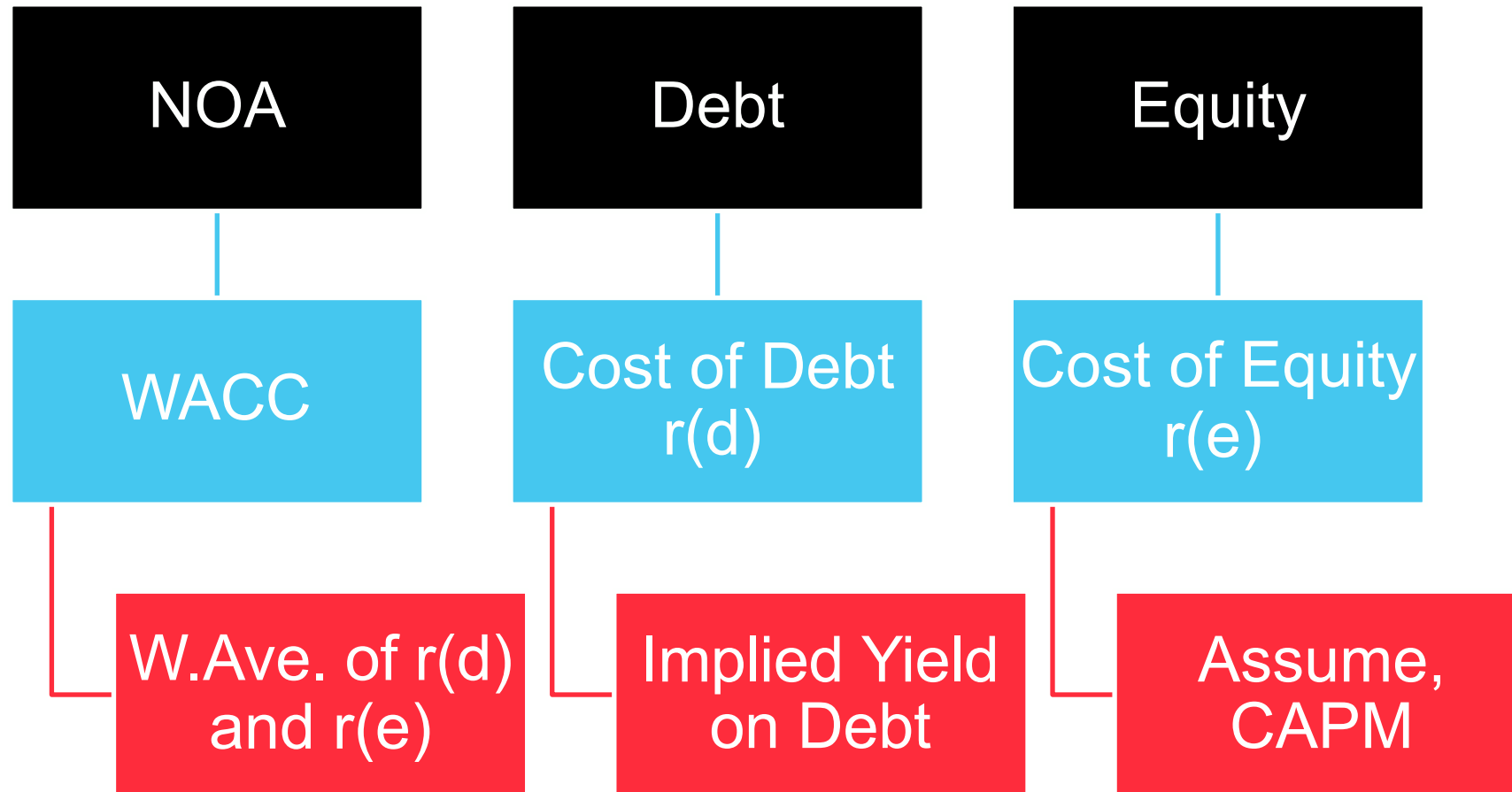
- **As we will discover in the seminars, the discount rate has a substantial effect on the resulting valuation.**
- **How do we determine the discount rate?**
- **In valuation models, this rate reflects an expected return (cost of capital) on funds**
 - **If I (1) took the cost/value it would cost me to buy the cash flows today, and (2) earned the expected return on it, (3) it would be equivalent to the future cash flow I would receive**
- **As we've discussed, this rate may be different depending on what we are valuing**



Determining Equity Cost of Capital

- The equity cost of capital proxies for the expected return on committing capital now to an equity investment
- This return can vary with a variety of factors.
- Majority of that variation comes with risk
 - Investors generally prefer to be compensated for taking great risk
- Accordingly, use a model that describes the return as a function of risk

Determining Cost of Capital





Cost of Equity

- **How can we determine the expected return or cost of capital for equity funding?**

Evaluate with CAPM (Capital Asset Pricing Model) – often used (market’s determination of expected return)

If we use CAPM, we are asking: “at what price will I receive the market’s return of X ” expected

Cost of equity capital

CAPM:

$$r_e = r_f + \beta \left[E(r_m) - r_f \right]$$

Market risk premium

Premium for “systematic” risk

Risk-free rate

r_f = risk free investment return (government bonds return)

r_m = return on the market portfolio of equities

Beta = relationship between expected returns and the premium $r_m - r_f$ to return for taking risk by investing in equities. Beta reflects *systematic* risk.

Beta, risk-free rate and market risk premium

➤ Risk-free rate:

- Rate on intermediate-term government bonds (incorporates future inflation effects)

➤ Market Risk Premium:

- Study focusing on the returns to an international portfolio
- Risk premium changes over time.

<http://www.market-risk-premia.com/au.html>

➤ Beta:

- Historical association between **firm-specific returns** and market returns
- relationship between expected returns and the premium $r_m - r_f$ for taking risk by investing in equities. Beta reflects *systematic risk*.



Cost of capital recap

➤ **Cost of debt (with a consideration of tax):**

- Should reflect the current interest rate(s)
- Must be **net of taxes** because after-tax cash flows are being discounted

➤ **Cost of equity:**

- A complex topic
- The CAPM provides one approach
- CAPM may be combined with firm size
- Amount of leverage affects risk



Determining Debt Cost of Capital



$$r_d = r_f + \text{debt risk premium}$$

- **Government debt is considered risk free. Corporate debt has risk.**
- **Risk is related to duration. Will longer or shorter debt have more risk for the debt holder?**
- **How would this vary with company financial “strength”?**

Determining Debt Cost of Capital

- **There are numerous challenges to determining debt cost of capital**
- **How can we get evidence as to the debt risk premium?**
 1. If has quoted bonds – look to bond yield
 2. If no quoted bonds – look to existing loan disclosures in annual report for yield
 3. Look to yields/rates of comparable companies with similar characteristics
 4. Average interest rate (interest expenses / interest-bearing debt)

Weighted average cost of capital

Cost of Equity

$$r_e = r_f + \beta [E(r_m) - r_f]$$

After Tax Cost of debt

$$r_d = r_{debt} \cdot (1 - T)$$

WACC

$$WACC = \frac{V_d}{V_d + V_e} r_{debt} (1 - T) + \frac{V_e}{V_d + V_e} r_e$$

V_d = Value of net debt

V_e = Value of equity

r_{debt} = cost of debt

r_e = cost of equity capital

T = the tax rate

Estimating Weighted Average Cost of Capital (WACC)

➤ The appropriate discount rate to value a company's **assets** (debt plus equity) is the WACC, which takes into account debt and equity sources of financing:

$$\text{WACC} = \frac{\text{Net debt}}{\text{Net debt} + \text{Equity}} (1 - \text{Tax rate}) \times \text{Cost of debt} + \frac{\text{Equity}}{\text{Net debt} + \text{Equity}} \text{Cost of equity}$$

➤ Cost of debt (with a consideration of tax) = (net) interest rate on (net) debt
= NIEAT/ND

➤ Adjust debt for the tax rate (as it provides a reduction in taxes)

➤ If preference shares present, adds a term to the calculation

Common fundamental valuation models

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DDM pros and cons

Advantages:

Easy concept: dividends are what shareholders get, so forecast them

Predictability: dividends are usually fairly stable in the short run so
dividends are easy to forecast (in the short run)

Disadvantages:

Relevance: dividends payout is not related to value

Forecast horizons: typically requires forecasts for long periods;
terminal values for shorter periods are hard to calculate with
any reliability: the proportion of dividend-paying companies
declines over time; **how do you value a company
that never pays dividends?**



The Discounted Cash Flow (DCF) Model

- Also derived from Dividend Discount Model (DDM)
- Based on the insight that dividends can be viewed as distributions of free cash flows to equity holders
- If all FCFE is paid out as dividends, dividends equal FCFE.



Free Cash Flow Models

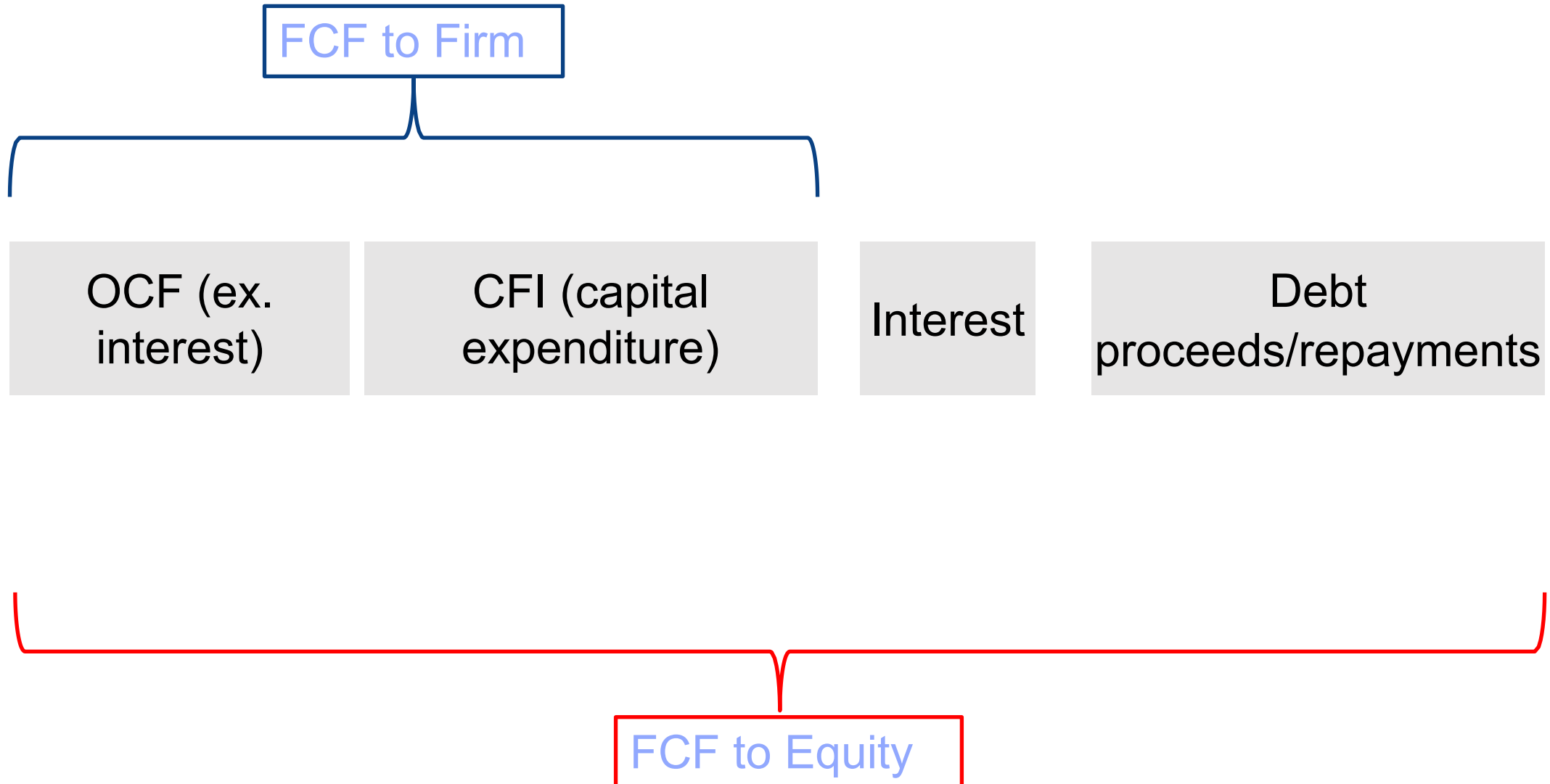
- Value cash earned by the company
- As discussed, can either value the equity directly (**levered** model) or the NOA (**unlevered** model)
- Should both these models come to identical results?



Free Cash Flow Models

- **Free cash flow for the purposes of these models:**
 - Levered model requires FCF available to shareholders
 - Levered model therefore requires FCF after interest, debt repayments (or proceeds) etc
 - Unlevered model requires FCF excluding all financing effects.
 - Unlevered model therefore requires FCF before interest and any debt trans.

Free Cash Flow Models



Free Cash Flow Models (Levered)

- Levered model in general (using FCF to Equity):

$$V_{Eq} = \sum_{t=1}^{\infty} \frac{FCFE_t}{(1 + k_+)^t} = \frac{FCFE_1}{(1 + k_+)^1} + \frac{FCFE_2}{(1 + k_+)^2} + \frac{FCFE_3}{(1 + k_+)^3} \cdots \frac{FCFE_T}{(1 + k_+)^T}$$

Free Cash Flow Models (Levered)

- In practice – for example:

$$V_{Eq} = \frac{FCFE_1}{(1 + k_e)^1} + \frac{FCFE_2}{(1 + k_e)^2} + \frac{FCFE_3}{(1 + k_e)^3} + \frac{TV}{(1 + k_e)^3}$$

$$TV = \frac{FCFE_4}{k_e - g_{FCFE}} = \frac{FCFE_3(1 + g)}{k_e - g_{FCFE}} = FCFE_4 \times P/FCFE \text{ Ratio}$$

↑
Option 1

↑
Option 2

↑
Option 3

Price to FCFE ratio: <https://www.alphaspread.com/security/nasdaq/aapl/relative-valuation/ratio/price-to-fcfe>

Free Cash Flow Models (Unlevered)

- **Unlevered model in general (using FCF to Firm):**

$$\begin{aligned} V_{NOA} &= \sum_{t=1}^{\infty} \frac{FCFF_t}{(1 + WACC)^t} \\ &= \frac{FCFF_1}{(1 + WACC)^1} + \frac{FCFF_2}{(1 + WACC)^2} + \frac{FCFF_3}{(1 + WACC)^3} \cdots \frac{FCFF_T}{(1 + WACC)^T} \end{aligned}$$

Free Cash Flow Models (Unlevered)

- In practice – for example:

$$V_{NOA} = \frac{FCFF_1}{(1 + WACC)^1} + \frac{FCFF_2}{(1 + WACC)^2} + \frac{FCFF_3}{(1 + WACC)^3} + \frac{TV}{(1 + WACC)^3}$$

$$TV = \frac{FCFF_4}{WACC - g_{FCFF}} = \frac{FCFF_3(1 + g)}{WACC - g_{FCFF}} = FCFF_4 \times EV / FCFF \text{ Ratio}$$

↑
Option 1

↑
Option 2

↑
Option 3



Free Cash Flow Models

- **Converting Value of NOA to Value of Equity:**
 - Need to subtract the Value of Net Debt.
 - Either value it separately, gather a market value, or assume book value

$$V_{NOA} = V_{Eq} + V_{Net\ Debt}$$

$\therefore$

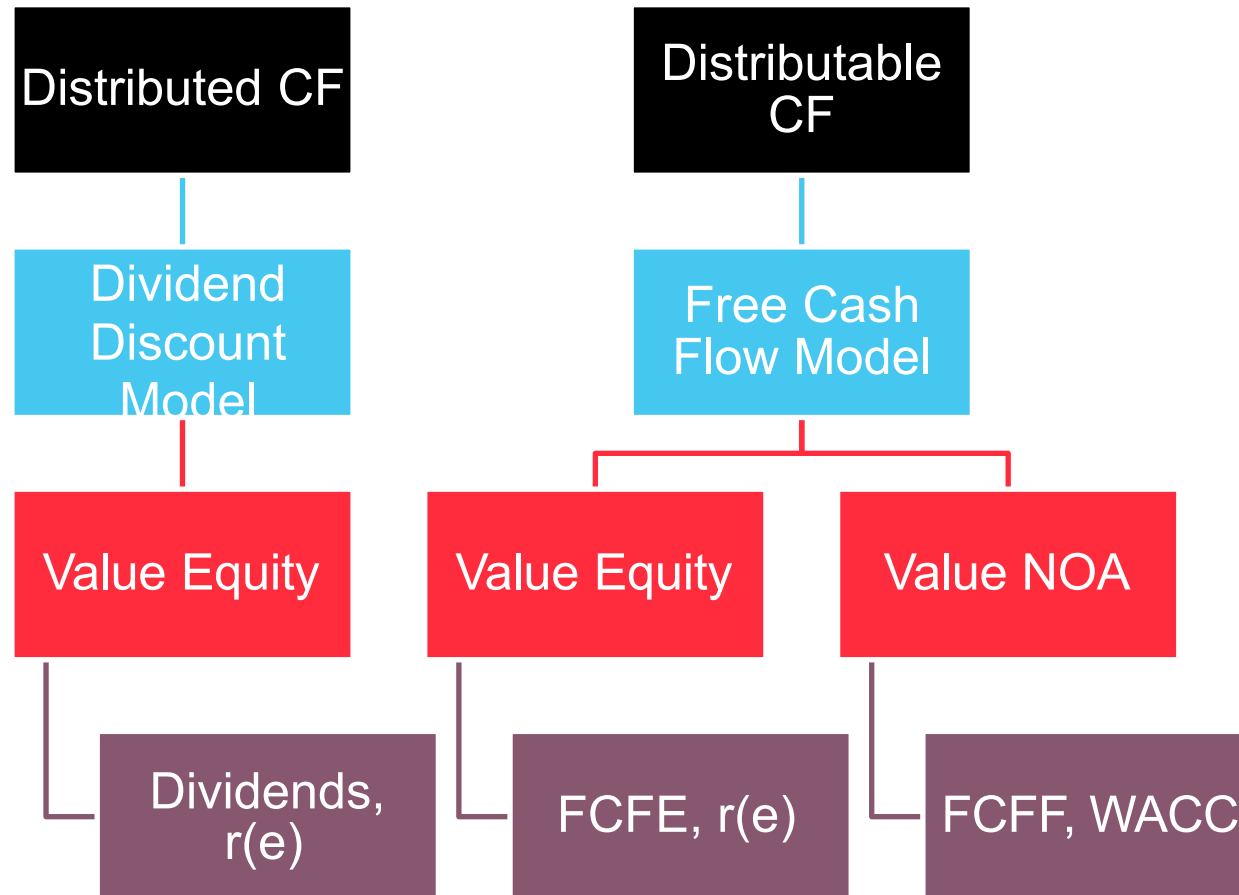
$$V_{Eq} = V_{NOA} - V_{Net\ Debt}$$



Growth rates

- Will have noted the presence of “g” in all models
- This is a growth rate into *perpetuity* – a long period of time
- When discussing dividends or free cash flow, what constraints are we likely to have on “g” and what do these mean?
 - $g < 0...$
 - $g = 0...$
 - $g > 0...$
 - $g > r...$

Summing Up





Conclusion

- **Final comments and takeaways**
 - Corporate valuations are based on the transfer of cash flows and the value those cash flows provide based on their “timing”
 - Valuation methods may be more or less appropriate for different types of businesses
 - Analysts should choose methods most suited to valuing particular businesses
 - Analysts can choose to value the NOA or the Equity of a business using different (but similar) approaches to valuation



Conclusion

- **How to practice**
 - See example valuations on the LMS
 - Value a set of forecasts (for example, Reddit [Seminar last week])
 - Work through the text examples
 - Practice valuations using all methods



Next week

- **We continue with:**
 - Residual income models
 - Relative valuation



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See you at the seminar

